# PROJECT REPORT

## Insta Mechanic – Smart On-Demand Vehicle Assistance Platform

### Submitted in Partial Fulfillment of the Requirements for the Degree of Bachelor of Technology in Computer Science & Engineering

### Submitted To

Department of Computer Science & Engineering
Dev Bhoomi Uttarakhand University

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### Academic Session

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# DECLARATION

We hereby declare that the project entitled **"Insta Mechanic – Smart On-Demand Vehicle Assistance Platform"** submitted to Dev Bhoomi Uttarakhand University in partial fulfillment of the requirements for the award of Bachelor of Technology in Computer Science & Engineering is an original work carried out by us.

This project has not been submitted previously to any other university or institution for the award of any degree or diploma. All sources of information used in this report have been properly acknowledged.

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# CERTIFICATE

This is to certify that the project report entitled **"Insta Mechanic – Smart On-Demand Vehicle Assistance Platform"** submitted by Kumar Raj Ranjan, Vihan Mudgal, and Umesh Roy has been completed under the guidance of the Department of Computer Science & Engineering, Dev Bhoomi Uttarakhand University.

The work presented in this report is genuine and fulfills the academic requirements for the completion of the Bachelor of Technology degree program.

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# ACKNOWLEDGEMENT

We express our heartfelt gratitude to the faculty members and mentors of the Department of Computer Science & Engineering, Dev Bhoomi Uttarakhand University, for their valuable guidance, encouragement, and technical support throughout the development of this project.

We are especially thankful to our project guide for providing continuous supervision and motivation during every phase of the project. Their suggestions helped us improve the overall quality and functionality of the application.

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# ABSTRACT


In today's rapidly evolving digital environment, the demand for quick and reliable vehicle assistance services has increased significantly. Vehicle owners frequently face situations such as breakdowns, battery failures, tyre punctures, and emergency roadside issues where immediate professional support becomes necessary. Traditional methods of searching for mechanics are often time-consuming, unreliable, and inefficient.

The proposed project, **Insta Mechanic**, is a smart web-based on-demand vehicle assistance platform designed to connect customers with nearby verified mechanics in real time. The platform provides an efficient and user-friendly system that allows customers to book services online, track mechanic status, and make secure payments.

The project aims to modernize roadside assistance by integrating technologies such as HTML5, CSS3, Bootstrap 5, JavaScript, Node.js, Express.js, PostgreSQL, Socket.io, and

payment gateway APIs. The system provides separate dashboards for customers, mechanics, and administrators, ensuring smooth communication and service management.

The platform focuses on improving customer convenience, reducing response time, and increasing service reliability. With future scalability and enhancement possibilities, Insta Mechanic can evolve into a fully integrated smart vehicle support ecosystem.

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# 1. INTRODUCTION

Digital transformation has changed the way services are accessed and delivered across industries. From food delivery to online transportation services, customers expect convenience, transparency, and instant accessibility. However, the roadside assistance and vehicle repair industry still relies heavily on traditional communication methods.

Vehicle breakdowns can occur unexpectedly and create serious inconvenience for users. In emergency situations, customers often struggle to locate trustworthy mechanics nearby. Delays in service availability can increase stress and negatively affect customer experience.

Insta Mechanic is designed as a modern digital platform that bridges the gap between vehicle owners and mechanics. The platform allows customers to request vehicle services instantly through a responsive web application.

The application provides:

- Real-time booking functionality
- Mechanic availability tracking
- Secure authentication
- Payment gateway integration
- Mechanic verification system
- Role-based dashboards
- Real-time service updates

The project combines frontend development, backend API integration, database management, authentication systems, and responsive UI design into a single scalable platform.

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# 2. EXISTING SYSTEM ANALYSIS

The current roadside assistance process is mostly manual and lacks digital automation.

## Problems in Existing Systems

### Manual Communication

Customers rely on phone calls or local references to find mechanics.

### Delayed Response Time

Finding a mechanic during emergencies can take a long time.

### Lack of Transparency

Customers cannot track booking progress or mechanic status.

### No Verification System

There is no centralized platform to verify mechanic authenticity.

### Payment Challenges

Most traditional services depend on cash payments.

### No Centralized Database

There is no proper system for storing booking history and service records.

The proposed system addresses these limitations through automation and digital integration.

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# 3. PROBLEM STATEMENT

Vehicle owners frequently encounter problems such as engine failure, battery issues, tyre punctures, fuel shortages, and emergency breakdowns. During such situations, users often face difficulties in finding nearby mechanics quickly.

The absence of a centralized digital platform creates several issues:

- Time-consuming mechanic search
- Unverified service providers
- Poor communication
- No live tracking system
- Lack of online booking management
- No integrated payment solution

These challenges reduce service efficiency and create inconvenience for users.

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# 4. OBJECTIVES OF THE PROJECT

The major objectives of Insta Mechanic are:

- To develop a responsive web application for vehicle assistance services
- To connect users with nearby verified mechanics
- To provide fast and reliable booking services
- To create separate dashboards for customers, mechanics, and administrators
- To implement secure authentication and authorization
- To enable real-time booking updates
- To integrate secure digital payment systems
- To improve transparency and customer satisfaction
- To maintain booking history and service records
- To create a scalable architecture for future expansion

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# 5. SCOPE OF THE PROJECT

The scope of the project includes:

## Customer Features

- Registration and login
- Service booking
- Live booking tracking
- Payment integration
- Booking history

## Mechanic Features

- Mechanic registration
- Availability management
- Accept or reject bookings
- Update service status
- Earnings tracking

## Admin Features

- Verify mechanics
- Monitor bookings
- Manage users
- Handle complaints
- Generate reports

Initially, the project targets one city or service area. Future versions can support multi-city operations and mobile applications.

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# 6. PROPOSED SOLUTION

Insta Mechanic provides a centralized web-based solution for managing vehicle assistance requests.

## Workflow of the Proposed System

1. Customer registers on the platform.
2. User selects the required service.
3. Booking details and location are submitted.
4. Nearby mechanics receive notifications.

5. Mechanic accepts the booking.
6. Booking progress is updated in real time.
7. Service is completed and payment is processed.
8. Admin monitors overall activities.

This workflow reduces service delays and improves customer convenience.

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# 7. FEASIBILITY STUDY

## Technical Feasibility

The technologies used in the project are open-source, modern, and widely supported.

## Economic Feasibility

The project can be developed with minimal infrastructure cost using cloud platforms and open-source frameworks.

## Operational Feasibility

The system is easy to use for customers, mechanics, and administrators.

## Legal Feasibility

The project follows standard privacy and security practices.

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# 8. LITERATURE SURVEY

Various online service platforms such as ride-sharing and food delivery systems inspired the development of Insta Mechanic.

Research shows that users prefer digital platforms because they provide:

- Faster service access
- Real-time updates
- Transparent pricing
- Secure payments
- Improved convenience

Modern web technologies such as Node.js, PostgreSQL, and Socket.io provide scalable solutions for developing such systems.

# 9. REQUIREMENT ANALYSIS

## Functional Requirements

- User authentication
- Booking management
- Mechanic availability tracking
- Payment integration
- Admin verification
- Real-time notifications

## Non-Functional Requirements

- High performance
- Responsive UI
- Security and privacy
- Scalability
- Reliability
- Maintainability

# 10. TECHNOLOGY STACK

| Layer | Technology |
|---|---|
| Frontend | HTML5, CSS3, Bootstrap 5, JavaScript |
| Backend | Node.js, Express.js |
| Database | PostgreSQL |
| Authentication | JWT, bcrypt |
| Realtime | Socket.io |
| Payments | Razorpay / Stripe |
| Hosting | Netlify, Vercel, Render, Railway |
| Testing | Jest, Playwright, Supertest |

## Why Bootstrap?

Bootstrap provides responsive UI components and simplifies mobile-first web development.

### Why Node.js?

Node.js offers fast asynchronous operations and scalability.

### Why PostgreSQL?

PostgreSQL ensures reliable relational database management.

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## 11. SYSTEM ARCHITECTURE

The project follows a three-tier architecture:

### Presentation Layer

Handles frontend UI and user interactions.

### Application Layer

Processes business logic and API communication.

### Database Layer

Stores users, bookings, services, and payments.

The architecture ensures modularity, maintainability, and scalability.

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## 12. FUNCTIONAL MODULES

### Customer Module

- User registration
- Login system
- Booking services
- Viewing service history
- Payment management

### Mechanic Module

- Accept bookings
- Update availability
- Manage earnings
- Update service progress

### Admin Module

- Verify mechanics
- Manage users
- Track payments
- View reports

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## 13. DATA FLOW DIAGRAM

### Level 0 DFD

The customer sends booking requests to the system, which forwards them to nearby mechanics.

### Level 1 DFD

The system processes authentication, booking creation, payment processing, and notification management.

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## 14. USE CASE DIAGRAM

### Customer Use Cases

- Register
- Login
- Book Service
- Track Booking
- Make Payment

### Mechanic Use Cases

- Accept Booking
- Update Status
- Toggle Availability

### Admin Use Cases

- Verify Mechanic
- Manage Users
- View Reports

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# 15. DATABASE DESIGN

The database design ensures proper normalization and relationship management.

## Users Table

Contains customer, mechanic, and admin details.

## Mechanics Table

Stores mechanic verification and location details.

## Services Table

Stores available vehicle services.

## Bookings Table

Maintains booking records and statuses.

## Payments Table

Stores payment information.

## Reviews Table

Maintains customer ratings and feedback.

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# 16. API DESIGN

REST APIs are used for frontend-backend communication.

### Authentication APIs

- POST /api/auth/register
- POST /api/auth/login

### Booking APIs

- POST /api/bookings
- GET /api/bookings/:id
- PATCH /api/bookings/:id/status

### Mechanic APIs

- GET /api/mechanics/nearby

- PATCH /api/mechanics/availability

### Payment APIs

- POST /api/payments/create-order
- POST /api/webhooks/payment

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## 17. FRONTEND DESIGN

The frontend is developed using responsive Bootstrap components.

### Key Pages

- Home Page
- Services Page
- Booking Page
- Mechanic Dashboard
- Admin Dashboard

### UI Components

- Navbar
- Cards
- Forms
- Tables
- Alerts
- Toasts
- Progress Bars

The design focuses on simplicity and accessibility.

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## 18. BACKEND DESIGN

The backend handles:

- Authentication
- Booking processing
- Database operations
- API communication
- Payment verification

- Real-time notifications

Express.js middleware is used for validation and security.

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# 19. AUTHENTICATION AND AUTHORIZATION

## JWT Authentication

JSON Web Tokens are used for secure session management.

## Password Security

Passwords are hashed using bcrypt.

## Role-Based Access Control

Different access permissions are assigned to:

- Customers
- Mechanics
- Administrators

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# 20. REAL-TIME COMMUNICATION

Socket.io is used to provide:

- Live booking updates
- Mechanic location tracking
- Instant notifications
- Real-time status synchronization

This improves communication efficiency.

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# 21. PAYMENT GATEWAY INTEGRATION

The system integrates Razorpay or Stripe for secure online transactions.

## Payment Features

- Secure payment processing

- Payment verification
- Booking receipt generation
- Transaction history

This ensures reliable financial transactions.

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# 22. SECURITY FEATURES

The project implements multiple security mechanisms:

- Password hashing
- JWT authentication
- Rate limiting
- Input validation
- HTTPS support
- Secure API communication
- CORS protection
- Helmet middleware

These features help protect the platform from common cyber threats.

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# 23. TESTING AND VALIDATION

## Unit Testing

Individual modules and functions are tested.

## Integration Testing

Frontend and backend integration is verified.

## System Testing

Complete workflow testing is performed.

## Performance Testing

Load testing ensures system stability.

### Security Testing

Authentication and validation mechanisms are verified.

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## 24. DEPLOYMENT STRATEGY

### Frontend Deployment

Hosted using Netlify or Vercel.

### Backend Deployment

Hosted using Render or Railway.

### Database Deployment

PostgreSQL hosted on cloud platforms.

### Monitoring

Analytics and uptime monitoring tools are integrated.

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## 25. PERFORMANCE OPTIMIZATION

Several techniques are used for optimization:

- Database indexing
- API caching
- Code minification
- Image optimization
- Lazy loading
- Efficient API handling

These techniques improve speed and scalability.

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## 26. ADVANTAGES OF THE SYSTEM

- Quick roadside assistance
- Easy booking management

- Verified mechanics
- Secure payments
- Live tracking system
- Responsive user interface
- Improved transparency
- Better customer experience

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## 27. LIMITATIONS OF THE SYSTEM

- Internet connection is required
- GPS accuracy may vary
- Initial service area is limited
- Real-time features increase server load
- Dependency on third-party payment APIs

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## 28. FUTURE ENHANCEMENTS

Future improvements include:

- Android and iOS applications
- AI-based mechanic recommendations
- Voice assistant support
- GPS live route navigation
- Subscription services
- Predictive vehicle maintenance
- Advanced analytics dashboard
- Multi-language support
- Smart chatbot integration

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## 29. CONCLUSION

Insta Mechanic is an innovative and practical solution for modern vehicle assistance management. The project successfully demonstrates the integration of frontend technologies, backend development, relational database systems, authentication mechanisms, payment gateways, and real-time communication.

The platform improves accessibility, minimizes response time, and enhances overall customer satisfaction. The system also provides business scalability and future enhancement opportunities.

The successful implementation of Insta Mechanic proves the effectiveness of digital transformation in the roadside assistance industry.

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# 30. REFERENCES


1. Node.js Official Documentation

2. Express.js Official Documentation

3. PostgreSQL Documentation

4. Bootstrap 5 Documentation

5. Socket.io Documentation

6. Razorpay API Documentation

7. JWT Authentication Documentation

8. HTML5 and CSS3 Standards

9. REST API Design Guidelines

10. Agile Software Development Methodology


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# APPENDIX

## Suggested Project Structure

insta-mechanic/

frontend/

- index.html
- services.html
- book.html

- about.html
- contact.html
- mechanic-dashboard.html
- admin/index.html

backend/

- src/index.js
- routes/
- controllers/
- models/
- middlewares/
- utils/

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